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B.Tech. Degree V Semester Supplementary Examination
April 2019

SE 15-1505 MANUFACTURING PROCESSES
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer **ALL** questions)

(10 × 2 = 20)

- I. (a) How are engineering materials classified?
 (b) Write a note on classification of steels.
 (c) List the purpose and methods of heat treatment of steels.
 (d) Distinguish between TIG and MIG welding.
 (e) How are resistance welding classified? Briefly explain any one of them.
 (f) List the defects in welding with remedies.
 (g) Name the various patterns which are normally used in foundry practice.
 (h) List various casting defects with remedies.
 (i) Briefly describe methods and process of rolling.
 (j) How would you classify extrusion process?



PART B

(4 × 10 = 40)

- II. Explain mechanical, thermal, chemical and technological properties of engineering materials. (10)

OR

- III. (a) Explain hardening process in heat treatment. (5)
 (b) Write short notes on Grey C.I. and white C.I. (5)

- IV. (a) How are welding process classified? (3)
 (b) Explain Electron Beam welding process with applications. (7)

OR

- V. (a) Write a note on solid state welding. (3)
 (b) With sketch explain HAZ. (7)

- VI. (a) Describe various pattern allowances used in casting. (5)
 (b) Explain investment casting process with merits, demerits and applications. (5)

OR

- VII. (a) Briefly explain the properties and classification of moulding sands. (7)
 (b) Write a note on slush casting. (3)

- VIII. (a) List various types of drilling machines and explain any one of them. (5)
 (b) With sketch explain AJM. Also describe its merits, demerits and applications. (5)

OR

- IX. (a) Write a note on classification of machine tools. (3)
 (b) With sketch explain EDM. Also describe its merits, demerits and applications. (7)